

A linear guard and a constant-width Gaussian solve

Downstream assembly refinements for integer multiplication

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Abstract

These notes change two downstream components of the conditional integer-multiplication bound, leaving both finite networks untouched. First, the coefficient guard of a simultaneous butterfly layer is bounded linearly in the number of axes, by charging each completed recursive child with its exact kernel instead of its internal dependency depth. The guard condition $\epsilon C_1 < 1$ becomes $\epsilon < 1$. Second, the Gaussian resampling system is solved by a precomputed banded LU factorization with a complete fixed-point error analysis. A diagonal similarity makes the system strongly diagonally dominant for every separation θ , so the Gaussian width can be the constant $\alpha = 2$. The Gaussian row of the cost table drops from power $\frac{3}{4} + \delta + \frac{5}{4}\epsilon$ to $\frac{1}{2} + \delta + \epsilon$. For certified savings a_b, a_c the parameter supremum moves from $Q/(5 + 4Q)$ to $Q/(2 + 2Q)$, $Q = \min(a_b, a_c)$. With the current certified savings, exact rational parameters give

$$T(n) = O(n(\log n)^{1-\kappa}), \quad \kappa = \frac{296461013}{2 \cdot 10^{17}} = 1.4823050650 \cdot 10^{-9} > 2^{-30},$$

about 2.5 times the previous conditional witness $5929220328/10^{19}$. The result is conditional on the same upstream interfaces as before, plus the new written proofs. The exact checks support those proofs; they do not independently establish the complete upstream theorem.

1 Baseline, dependencies, and the change

The baseline is odepoint's *intmultspam* [1], revision 8e1701f, a fork of Douglas Colkitt's *integer-mult-bounds* [2]. Its conditional witness is $\kappa = 5929220328/10^{19} > 2^{-31}$, which is the exact parameter supremum $a_b/(5 + 4a_b)$ of its assembly for the certified bit saving $a_b = 29646101715649/10^{22}$. Both rest on OpenAI's *Integer multiplication below $n \log n$* [3], pinned at

adc7f1241b42e322a6451854ab7e4b4c146bf78a.

Numbered citations to Harvey and van der Hoeven [4] refer to its author-hosted manuscript, as in the pinned source.

We retain every interface that the baseline assumes: the pinned stream operations, bit transfer and complex phase transfer, the paired bit network and the pair-star complex network with their certified savings $a_b = 29646101715649/10^{22}$ and $a_c = 4/10^9$, nonadjacent routing, compact dirty controls with their reservations and local repair, synthetic transforms, prime selection, the Gaussian identity and its forward and diagonal maps, and the final rounding. Two components change.

1. The width guard of the simultaneous layer (Section 2). The retained depth recurrence $A(e) \leq sA(e/m) + E$ gives widths $p + O(d^{C_1})$ with $C_1 = 5 - 4\beta + \zeta$. We prove widths $p + O(d)$.

*Prepared with AI assistance (Claude agents). This note builds on odepoint's *intmultspam*, a fork of Douglas Colkitt's *integer-mult-bounds*, and on OpenAI's paper *Integer multiplication below $n \log n$* . It has no independent review or endorsement from their authors.

2. The solve of the square Gaussian system (Section 3). The retained Neumann series costs $O(tp^{5/2+\delta}/(\alpha^3\theta))$ on a line of length t and is used under the hypothesis $\theta > p/\alpha^4$, so that it does not dominate the $O(tp^{3/2+\delta}\alpha)$ forward map [4, Section 4.4.2]. The retained separations are $\theta_i \asymp 1/d$, so $\alpha \asymp (pd)^{1/4}$, and the Gaussian row has power $\frac{3}{4} + \delta + \frac{5}{4}\epsilon$: one factor d for the axes and $\alpha \asymp p^{(1+\epsilon)/4}$. Harvey and van der Hoeven remark that a precomputed LU decomposition “allows us to relax the condition $\theta > p/\alpha^4$ to merely $\theta > 1/\alpha^2$ ”, and that they did not use it “because the error analysis is considerably more intricate than for the iterative method” [4, Section 4.4.2, pp. 34–35]. With $\theta_i \asymp 1/d$ that condition would still force $\alpha \asymp \sqrt{d}$. We prove a complete version that needs neither condition, for every integer $\alpha \geq 2$, and use $\alpha = 2$.

	Baseline	This note
Coefficient guard	$\epsilon(5 - 4\beta + \zeta) < 1$	$\epsilon < 1$
Gaussian width α	$\lceil (32db)^{1/4} \rceil$	2
Gaussian row power	$\frac{3}{4} + \delta + \frac{5}{4}\epsilon$	$\frac{1}{2} + \delta + \epsilon$
Source scale γ	$O(p^{1/2+3\epsilon/2})$	$O(d^2) = O(p^{2\epsilon})$
Parameter supremum	$Q/(5 + 4Q)$	$Q/(2 + 2Q)$
Witness κ	$5.929220328 \cdot 10^{-10}$	$1.4823050650 \cdot 10^{-9}$

2 Coefficient widths of the simultaneous layer

The simultaneous layer applies $C^{\otimes D}$, with $C = aI + bX$, $a = (1+i)/2$, $b = (1-i)/2$, to $D \leq d$ selected address bits, and converts it to the normalized butterfly layer by $H_0^{\otimes D} = b^D \mathcal{S}_D C^{\otimes D} \mathcal{S}_D$, where $\mathcal{S}_D$ multiplies each record by i to the number of selected bits equal to one. Its recursion splits a group of $e = mf$ axes into m slots, applies the finite pair-star network with W roles and s child calls, stops below d^β axes, and uses individual kernels at the leaves and on at most d reserved axes. The node count (PS11a)–(PS11b) cited below is the one in `notes/complex-pair-star-construction.tex`, which the pair-star manuscript patch `patches/complex-pair-star-31.patch` inserts into Section 3: it gives $D_{\text{scalar}} = 3v^2(12R_{\text{side}} + 4vh + 10v) + 8W$ for the scalar circuit and its endpoint and bank corrections, and $D_{\text{node}} = D_{\text{scalar}} + 4s$, where $4s$ pays for the inverse-child wrappers. The following subsection replaces the guard subsection of the retained layer proof.

2.1 A linear coefficient guard

The accelerated layer is evaluated exactly until the entire normalized butterfly layer is complete, and its intermediate operators need not be contractions. We bound the denominators and moduli of every intermediate coefficient by a quantity that is *linear* in d . The bound replaces the dependency-depth recurrence $A(e) \leq sA(e/m) + E$, which charges each of the s consecutive children of a node with the whole internal depth of that child, and therefore grows like a fixed power of d . Here a *completed* child is charged only for its exact action, which is a tensor power of the fixed kernel C .

Use the complex network’s own constants $m \geq 3$, W and s , independently of the bit network used by swaps. One network invocation is the computation at one internal node: its scalar gates, endpoint and bank corrections, inverse-child wrappers, binary changes of basis, row splitting and reassembly, and exactly s child calls. Let D_{node} bound the number of elementary coefficient operations of one invocation outside its child calls; an elementary operation is a complex addition, subtraction, halving, sign change or multiplication by i , applied coefficientwise to aligned records. For the pair-star network the finite count (PS11a)–(PS11b) of the pair-star construction gives

$$D_{\text{node}} < E = 64(W + m + 1)^3.$$

Binary changes of basis, compact-control swaps and rotations, exceptional repair, row padding, parking on the stack, copying and bank exchanges permute or copy complete coefficient encodings, or write zero rows; they perform no coefficient arithmetic.

For an integer $P \geq 0$ and real $M > 0$, call a finite family of complex numbers (P, M) -bounded if all its members lie in $2^{-P}\mathbb{Z}[i]$ and have modulus at most M . At a given moment, the *state family* of a call is the family of all coefficient values in that call's streams, including scratch roles, padded zero rows and parked streams.

Lemma 1. *If the state family is (P, M) -bounded and one elementary operation is applied coefficientwise, the new state family is $(P + 1, 2M)$ -bounded. Permutations and copies of complete coefficient encodings, and insertion of zero rows, preserve (P, M) -boundedness.*

Proof. The set $2^{-P}\mathbb{Z}[i]$ is closed under sums, differences, sign changes and multiplication by i , and $|a \pm b| \leq 2M$; halving maps it into $2^{-P-1}\mathbb{Z}[i]$ and does not increase moduli. Values that are not changed keep their bounds, and a (P, M) -bounded family is $(P + 1, 2M)$ -bounded. Movement creates no new values, and 0 lies in every such family. $\square$

Lemma 2. *Let $f \geq 1$ and let $\mathcal{K}$ be $C^{\otimes f}$ or $(C^{-1})^{\otimes f}$, acting on f selected address bits of a stream, coefficientwise and as the identity on all other address bits. If the values of the stream are (P, M) -bounded, then those of $\mathcal{K}$ applied to the stream are $(P + \lceil f/2 \rceil, 2^{f/2}M)$ -bounded.*

Proof. Put $a = (1 + i)/2$ and $b = (1 - i)/2 = -ia$. For $x, y \in \mathbb{F}_2^f$ with k the Hamming weight of $x + y$, the entry of $C^{\otimes f}$ at (x, y) is $a^{f-k}b^k = (-i)^k a^f = (-i)^k (1 + i)^f 2^{-f}$, and the entry of $(C^{-1})^{\otimes f}$, with $C^{-1} = bI + aX$, is $b^{f-k}a^k = (-i)^{f-k} a^f$. Since $(1 + i)^2 = 2i$, each entry is a unit times $(1 + i)^{f \bmod 2} 2^{-\lceil f/2 \rceil}$: it lies in $2^{-\lceil f/2 \rceil}\mathbb{Z}[i]$ and has modulus $2^{-f/2}$. An output coefficient is a $\mathbb{Z}[i]$ -combination of 2^f input coefficients with such entries. Hence it lies in $2^{-P-\lceil f/2 \rceil}\mathbb{Z}[i]$, and the maximum absolute row sum $2^f \cdot 2^{-f/2}$ bounds its modulus by $2^{f/2}M$. $\square$

For a node with e active axes, an integer Γ is a *valid growth* if, whenever the node's input family is (P, M) -bounded, every value occurring during its execution, including inside all descendants, is $(P + \Gamma, 2^\Gamma M)$ -bounded.

Lemma 3. *A leaf with $e < d^\beta$ active axes has valid growth $8e$. If every child of an internal node with $e = mf$ active axes has valid growth Γ_f , then*

$$\Gamma_e = D_{\text{node}} + s\lceil f/2 \rceil + \Gamma_f$$

is a valid growth for the node. A completed node with e active axes maps a (P, M) -bounded input family to a $(P + \lceil e/2 \rceil, 2^{e/2}M)$ -bounded output family.

Proof. A leaf applies e individual kernels, each with at most eight elementary operations; apply Lemma 1 at most $8e$ times.

At an internal node, apart from movement, the invocation performs at most D_{node} elementary operations and exactly s child calls on f axes. A completed forward child acts on its role stream as $C^{\otimes f}$ on the selected bits of its slot and as the identity elsewhere. An inverse child is one forward child between the unit wrappers $Z^{\otimes f}$ and the scalar $(-i)^f$, so its completed action is $(C^{-1})^{\otimes f}$; the wrappers are among the counted elementary operations. By Lemmas 1 and 2, just before the j -th child call the state family is $(P + a_j, 2^{a_j}M)$ -bounded with $a_j \leq D_{\text{node}} + (j - 1)\lceil f/2 \rceil$, because $f/2 \leq \lceil f/2 \rceil$. The child's input is a subfamily and the parent's parked values are unchanged during the call, so every value during the call is within $P + a_j + \Gamma_f$ and $2^{a_j + \Gamma_f}M$. After all steps the bound is $D_{\text{node}} + s\lceil f/2 \rceil$, which proves the second claim.

Every completed invocation applies $C^{\otimes e}$ to every row of every role, including roles whose initial scratch values were arbitrary, and padded zero rows remain zero. Lemma 2 with $f = e$ gives the last claim. $\square$

Proposition 1 (Linear guard). *Put*

$$C_0^* = E + 2s + m + 26, \quad \Delta = C_0^* d.$$

For every input of the simultaneous layer whose coefficients are disk values in $2^{-p}\mathbb{Z}[i]$, and for every $1 \leq D \leq d$ and fixed rational $0 < \beta < 1$, every intermediate coefficient lies in $2^{-(p+\Delta)}\mathbb{Z}[i]$ and has modulus at most 2^Δ . Thus $p + \Delta$ fractional bits and $\Delta + 3$ integer and sign bits per real component represent every intermediate value exactly. If $d \leq b_d p^\epsilon$ with a fixed $0 < \epsilon < 1$, then $\Delta \leq p$ beyond a fixed cutoff, so these widths are $O(p)$.

Proof. The input family is $(p, 1)$ -bounded. The layer first applies at most d individual kernels on its reserved row, front and back axes, or on all $D \leq d$ axes in the small- D fallback; by Lemma 1 all values stay within $p + 8d$ and 2^{8d} .

The remaining active axes form $n \leq (m-1)(1 + \lceil \log_m d \rceil) \leq (m-1)d$ base- m pieces of sizes $e_1, \dots, e_n$ with $\sum_j e_j \leq d$, executed one after another. For a piece of size e_0 with internal sizes $e_i = e_0/m^i$ and leaf size $e_k < d^\beta$, Lemma 3 gives, using $\lceil x \rceil \leq x + 1$, $k \leq \lceil \log_m d \rceil \leq d$ and $e_0 \leq d$,

$$\Gamma_{e_0} \leq \sum_{i=1}^k (D_{\text{node}} + s \lceil e_i/2 \rceil) + 8e_k \leq d(D_{\text{node}} + s) + \frac{sd}{2(m-1)} + 8d.$$

When piece j completes, the family bound has grown by at most $\lceil e_j/2 \rceil$ in the denominator exponent and $e_j/2$ in the binary logarithm of the modulus, by the last claim of Lemma 3. Hence during piece j every value is within $p + 8d + \sum_{i < j} \lceil e_i/2 \rceil + \Gamma_{e_j}$, and after all pieces the family is within $p + 8d + d/2 + n$.

The two outer diagonals $\mathcal{S}_D$ are unit multiplications. The scalar $b^D = (-i/2)^{tb^{D \bmod 2}}$, $t = \lfloor D/2 \rfloor$, consumes $t \leq d/2$ fractional bits through the shift by t , which only lowers moduli, and at most nine further elementary operations. Altogether every intermediate value is within

$$d\left(D_{\text{node}} + s + \frac{s}{2(m-1)} + m + 16\right) + 9 \leq d(D_{\text{node}} + 2s + m + 25) \leq C_0^* d,$$

using $s/(2(m-1)) \leq s$, $9 \leq 9d$ and $D_{\text{node}} < E$. The modulus exponent obeys the same bound, because each step above increases it by at most the amount charged to the denominator exponent. A value in $2^{-(p+\Delta)}\mathbb{Z}[i]$ of modulus at most 2^Δ has real and imaginary numerators of absolute value at most $2^{p+2\Delta}$, so $p + \Delta$ fractional and $\Delta + 3$ integer and sign bits hold it exactly. Finally $C_0^* d \leq C_0^* b_d p^\epsilon \leq p$ for all large p when $\epsilon < 1$. $\square$

All compact-control operations, exceptional repair, padding and parking were counted as movement. The individually processed axes are disjoint and number at most d , which is the same $8d$ allowance used above. Thus the condition $\epsilon C_1 < 1$ of the dependency-depth guard is replaced by $\epsilon < 1$, and the stopping exponent β no longer enters any width condition.

3 Gaussian resampling with a constant width

Here $\mathbb{D}_p$ is the disk grid $2^{-p}\mathbb{Z}[i] \cap \{|z| \leq 1\}$, vectors carry the maximum coordinate norm, and an approximation $\tilde{L}$ of a linear contraction L has error E if $2^p \|\tilde{L}u - Lu\| < E$ for every input on the grid. The maps $\mathsf{S}, \mathsf{T}, \mathsf{C}, \mathsf{D}$ are those of [4, Section 4]: for $0 \leq k < t$,

$$(\mathsf{S}u)_k = \frac{1}{\alpha} \sum_{j \in \mathbb{Z}} e^{-\pi \alpha^{-2}(j-sk/t)^2} u_j, \quad (\mathsf{T}u)_k = \sum_{j \in \mathbb{Z}} e^{-\pi \alpha^2(tj/s-k)^2} u_j,$$

C selects the coordinates q_ℓ and $D = \text{diag}(e^{\pi\alpha^2\beta_\ell^2})$, so that $N = CTD$.

3.1 A precomputed banded solve

The cited one-dimensional resampling solves its square Gaussian system by a Neumann series [4, Lemma 4.12], which needs $\theta > p/\alpha^4$. Its Section 4.4.2 remarks that a precomputed LU decomposition would only need $\theta > 1/\alpha^2$, but the authors did not use it “because the error analysis is considerably more intricate”. We prove a precomputed banded solve with a complete fixed-point error analysis. With a diagonal rescaling it works for *every* θ and every $\alpha \geq 2$; it is therefore used below with the constant width $\alpha = 2$.

Throughout this subsection let $s < t < 2s$ be coprime positive integers, put $\theta = t/s - 1 \in (0, 1)$, and let $\alpha \geq 2$. For $\ell \in \mathbb{Z}$ write

$$x_\ell = \frac{t\ell}{s}, \quad q_\ell = \left\lfloor x_\ell + \frac{1}{2} \right\rfloor, \quad \beta_\ell = x_\ell - q_\ell \in \left[-\frac{1}{2}, \frac{1}{2}\right), \quad \Delta q_\ell = q_{\ell+1} - q_\ell,$$

and, for $h \in \mathbb{Z}$,

$$Q_h(\ell) = \left(\frac{th}{s} + \beta_\ell\right)^2 - \beta_{\ell+h}^2.$$

For $0 \leq \ell < s$ these are the selected coordinates q_ℓ and fractional parts β_ℓ of the cited Gaussian system, and $\beta_{\ell+s} = \beta_\ell$. The square matrix $N = CTD$ of the cited Section 4.2, written with indices modulo s , has entries

$$N_{\ell,j} = \sum_{h \equiv j-\ell \pmod{s}} e^{-\pi\alpha^2 Q_h(\ell)},$$

and $N = I + E$, where E collects all terms with $h \neq 0$; every such term is a positive real number. For matrices, $\|\cdot\|$ is the maximum absolute row sum, induced by the maximum coordinate norm.

Lemma 4 (Diagonally dominant LU factors). *Let M be a real $n \times n$ matrix with $|M_{ii} - 1| + \sum_{j \neq i} |M_{ij}| \leq \rho$ for every i , where $0 \leq \rho \leq \frac{1}{10}$. Gaussian elimination without pivoting runs to completion and gives $M = LU$, with L unit lower triangular and U upper triangular. Every Schur complement S in the elimination satisfies $S_{ii} \in [1 - \rho, 1 + \rho]$ and $\sum_{j \neq i} |S_{ij}| \leq \rho$ in each row. Moreover, for every i ,*

$$U_{ii} \in [1 - \rho, 1 + \rho], \quad \sum_{j > i} |U_{ij}| \leq \rho, \quad \sum_{j < i} |L_{ij}| \leq \frac{\rho}{1 - 2\rho} \leq \frac{1}{8},$$

and, with D_U the diagonal of U ,

$$\|L^{-1}\| \leq \frac{8}{7}, \quad \|(D_U^{-1}U)^{-1}\| \leq \frac{9}{8}, \quad \|U^{-1}\| \leq \frac{5}{4}, \quad \|M^{-1}\| \leq \frac{10}{9}.$$

Proof. For a matrix X with positive diagonal write $\omega_i(X) = \sum_{j \neq i} |X_{ij}|$, $\mu_i(X) = X_{ii} - \omega_i(X)$ and $\tau_i(X) = X_{ii} + \omega_i(X)$. Initially $\mu_i(M) \geq 1 - \rho > 0$ and $\tau_i(M) \leq 1 + \rho$. One elimination step with pivot 1 forms $X'_{ij} = X_{ij} - X_{i1}X_{1j}/X_{11}$ for $i, j \geq 2$. Then

$$\omega_i(X') \leq \omega_i(X) - |X_{i1}| + \frac{|X_{i1}|}{X_{11}}(\omega_1(X) - |X_{1i}|), \quad |X'_{ii} - X_{ii}| \leq \frac{|X_{i1}||X_{1i}|}{X_{11}},$$

hence $\mu_i(X') \geq \mu_i(X) + |X_{i1}|\mu_1(X)/X_{11} \geq \mu_i(X)$ and $\tau_i(X') \leq \tau_i(X) - |X_{i1}|\mu_1(X)/X_{11} \leq \tau_i(X)$. By induction every Schur complement S has positive pivots, $\mu_i(S) \geq 1 - \rho$ and $\tau_i(S) \leq 1 + \rho$; thus $S_{ii} \in [1 - \rho, 1 + \rho]$ and $\omega_i(S) = (\tau_i(S) - \mu_i(S))/2 \leq \rho$. Row k of U is row k of the k -th Schur complement $S^{(k)}$, which gives the bounds on U .

For L , fix i and put $\omega^{(k)} = \sum_{j \geq k, j \neq i} |S_{ij}^{(k)}|$. Eliminating pivot $k < i$ removes $|S_{ik}^{(k)}|$ from row i and adds at most $(|S_{ik}^{(k)}|/S_{kk}^{(k)})\omega_k(S^{(k)}) \leq |S_{ik}^{(k)}|\rho/(1 - \rho)$ to its other off-diagonal entries. Hence

$|S_{ik}^{(k)}|(1-2\rho)/(1-\rho) \leq \omega^{(k)} - \omega^{(k+1)}$, and summing over $k < i$ gives $\sum_{k < i} |S_{ik}^{(k)}| \leq \rho(1-\rho)/(1-2\rho)$. As $L_{ik} = S_{ik}^{(k)}/S_{kk}^{(k)}$ and $S_{kk}^{(k)} \geq 1-\rho$, the row bound on L follows. Finally $L-I$, $D_U^{-1}U-I$ and $M-I$ have norms at most $\frac{1}{8}$, $\frac{\rho}{1-\rho} \leq \frac{1}{9}$ and $\frac{1}{10}$, and $\|D_U^{-1}\| \leq \frac{10}{9}$; Neumann series give the four norm bounds. $\square$

The matrix N itself need not be diagonally dominant: when $\beta_{\ell+1}$ is close to $-\frac{1}{2}$, the entry $N_{\ell,\ell+1}$ is close to one. A diagonal similarity removes this. Let ψ be the 1-periodic function with

$$\theta\psi(x) = \begin{cases} x(x + \frac{1}{2}), & -\frac{1}{2} \leq x \leq -\frac{1}{4}, \\ -\frac{1}{16}, & -\frac{1}{4} \leq x \leq \frac{1}{4}, \\ x(x - \frac{1}{2}), & \frac{1}{4} \leq x \leq \frac{1}{2}, \end{cases}$$

and put

$$\Phi = \frac{1}{16\theta}, \quad \varphi_\ell = \psi(\beta_\ell) + \Phi \in [0, \Phi], \quad \Lambda = \text{diag}(e^{\pi\alpha^2\varphi_\ell})_{\ell \in \mathbb{Z}/s}, \quad R = e^{\pi\alpha^2\Phi}.$$

Thus every diagonal entry of Λ lies in $[1, R]$.

Lemma 5 (Scaled dominance). *For every ℓ and $h \neq 0$ the scaled exponent $\mathcal{E}_{\ell,h} = Q_h(\ell) - (\varphi_{\ell+h} - \varphi_\ell)$ satisfies*

$$\mathcal{E}_{\ell,\pm 1} \geq \frac{1}{2} + \theta, \quad \mathcal{E}_{\ell,h} \geq |h|(|h| - \frac{3}{2}) \quad (|h| \geq 2).$$

Consequently $\Lambda^{-1}N\Lambda = I + \Lambda^{-1}E\Lambda$ with

$$\|\Lambda^{-1}E\Lambda\| \leq \rho_\alpha := 2e^{-\pi\alpha^2/2} + 2.01e^{-\pi\alpha^2} \leq \rho_2 < 0.0038,$$

N is invertible with $\|N^{-1}\| \leq R/(1-\rho_2) \leq 1.004R$, and for every integer $m \geq 1$ the terms with $|h| > m$ contribute less than $3e^{-\pi\alpha^2 m^2}$ to each row of $\Lambda^{-1}E\Lambda$.

Proof. (a) Since $x_{\ell+1} - x_\ell = 1 + \theta \in (1, 2)$ and $|\beta| \leq \frac{1}{2}$, we have $\Delta q_\ell \in \{1, 2\}$ and $\beta_{\ell+1} = \beta_\ell + \theta + 1 - \Delta q_\ell$.

(b) As $t/s + \beta_\ell = x_{\ell+1} - q_\ell = \beta_{\ell+1} + \Delta q_\ell$ and $-t/s + \beta_{\ell+1} = x_\ell - q_{\ell+1} = \beta_\ell - \Delta q_\ell$,

$$Q_1(\ell) = \Delta q_\ell(\Delta q_\ell + 2\beta_{\ell+1}), \quad Q_{-1}(\ell+1) = \Delta q_\ell(\Delta q_\ell - 2\beta_\ell).$$

(c) For $h \neq 0$, $|th/s + \beta_\ell| \geq (t/s)|h| - \frac{1}{2} > |h| - \frac{1}{2} \geq \frac{1}{2}$, so $Q_h(\ell) > (|h| - \frac{1}{2})^2 - \frac{1}{4} = |h|(|h| - 1) \geq 0$, as in the proofs of [4, Lemmas 4.6 and 4.11].

(d) We have $\psi(x) = \theta^{-1} \int_{-1/2}^x g$, where $g(y) = 2y + \frac{1}{2}$ on $[-\frac{1}{2}, -\frac{1}{4}]$, $g = 0$ on $[-\frac{1}{4}, \frac{1}{4}]$ and $g(y) = 2y - \frac{1}{2}$ on $[\frac{1}{4}, \frac{1}{2}]$. On $[-\frac{1}{2}, \frac{1}{2}]$ one has $2y - \frac{1}{2} \leq g(y) \leq 2y + \frac{1}{2}$ and $|g(y)| \leq \frac{1}{2}$, and $\int_{-1/2}^{1/2} g = 0$; so ψ is continuous and 1-periodic, and $\theta\psi$ takes values in $[-\frac{1}{16}, 0]$.

(e) In both cases of (a), periodicity gives $\varphi_{\ell+1} - \varphi_\ell = \psi(\beta_\ell + \theta) - \psi(\beta_\ell) = \theta^{-1} \int_{\beta_\ell}^{\beta_\ell + \theta} \tilde{g}$, where $\tilde{g}$ is the periodic extension of g . Hence $|\varphi_{\ell+1} - \varphi_\ell| \leq \frac{1}{2}$ for every ℓ . If $\Delta q_\ell = 1$, then $[\beta_\ell, \beta_\ell + \theta] \subset [-\frac{1}{2}, \frac{1}{2})$, and averaging the bounds of (d) gives

$$2\beta_\ell + \theta - \frac{1}{2} \leq \varphi_{\ell+1} - \varphi_\ell \leq 2\beta_\ell + \theta + \frac{1}{2}.$$

(f) If $\Delta q_\ell = 1$, then by (b) $\mathcal{E}_{\ell,1} = 1 + 2\beta_\ell + 2\theta - (\varphi_{\ell+1} - \varphi_\ell) \geq \frac{1}{2} + \theta$ and $\mathcal{E}_{\ell+1,-1} = 1 - 2\beta_\ell + (\varphi_{\ell+1} - \varphi_\ell) \geq \frac{1}{2} + \theta$. If $\Delta q_\ell = 2$, then $Q_1(\ell) \geq 2$ and $Q_{-1}(\ell+1) \geq 2$, and both scaled exponents are at least $\frac{3}{2}$.

(g) For $|h| \geq 2$, (e) gives $|\varphi_{\ell+h} - \varphi_\ell| \leq |h|/2$, and (c) gives $\mathcal{E}_{\ell,h} \geq |h|(|h| - 1) - |h|/2$.

(h) Row ℓ of $\Lambda^{-1}E\Lambda$ is the sum, over all $h \neq 0$, of the positive terms $e^{-\pi\alpha^2\mathcal{E}_{\ell,h}}$; the terms with $h \equiv 0 \pmod{s}$ lie on the diagonal. By (f) and (g) the total is at most $2e^{-\pi\alpha^2/2} + 2\sum_{k \geq 2} e^{-\pi\alpha^2 k(k-3/2)}$,

and successive terms of the last sum decrease by at least $e^{-14\pi}$ when $\alpha \geq 2$. This gives ρ_α , which decreases in α ; the numerical bound $\rho_2 < 0.0038$ is an exact rational comparison. Neumann's series bounds $\|(I + \Lambda^{-1}E\Lambda)^{-1}\|$ by $1/(1 - \rho_2) < 1.004$, and $N^{-1} = \Lambda(I + \Lambda^{-1}E\Lambda)^{-1}\Lambda^{-1}$. For the tail, $k(k - \frac{3}{2}) \geq (m + 1)(m - \frac{1}{2}) \geq m^2$ for $k \geq m + 1 \geq 2$, so the terms with $|h| > m$ total less than $2.01e^{-\pi\alpha^2m^2}$. $\square$

We now use an integer α with $2 \leq \alpha < \sqrt{p}$, where $p > 100$ and $t < 2^p$. Put

$$L = \left\lceil \frac{4533\alpha^2s}{16000(t-s)} \right\rceil, \quad w = 3p + 2L + 12, \quad m = \left\lceil \frac{\sqrt{w}}{\alpha} \right\rceil,$$

$$\mathcal{B} = \{s - m, \dots, s - 1\}, \quad J_k = ((k, k + m] \cup \mathcal{B}) \cap (k, s),$$

$$\mathcal{P} = \{(i, j) : |i - j| \leq m\} \cup (\mathcal{B} \times [0, s)) \cup ([0, s) \times \mathcal{B}),$$

with indices in $[0, s)$, and assume $L \leq p$ and $s \geq 4m$. Since $\pi \log_2 e < 4.533$ and $s/(t - s) = 1/\theta$, we have $\log_2 R \leq L$. Write Q_w for truncation toward zero of the real and imaginary parts to $2^{-w}\mathbb{Z}$, so that $|Q_w z - z| < \sqrt{2}2^{-w}$ and $|Q_w z| \leq |z|$.

Lemma 6 (Precomputed banded solve). *Under these hypotheses put $J = N^{-1}/2^{L+2}$, so that $\|J\| \leq 0.251$.*

1. *A fixed multitape algorithm, given p, s, t, α , writes a factor stream of $O(sm w)$ bits in time $O(sm^2 w^2)$.*
2. *A fixed multitape algorithm, given that stream and $u \in \mathbb{D}_p^s$ in the scalar component format and increasing coordinate order, returns $\tilde{J}u \in \mathbb{D}_p^s$ in the same format with $2^p \|\tilde{J}u - Ju\| < 2$, in time $O(sm w^{1+\delta})$.*

The tape count and the implicit constants are independent of p, s, t, α .

Proof. Write $X_\Lambda = \Lambda^{-1}X\Lambda$ for matrices and $y_\Lambda = \Lambda^{-1}y$ for vectors. The machine never uses Λ ; it appears only in the analysis. As Λ has diagonal entries in $[1, R]$ and $R \leq 2^L$, an entrywise perturbation of size η has scaled entries at most $2^L\eta$, and $\|y_\Lambda\| \leq \|y\| \leq 2^L\|y_\Lambda\|$.

Truncation. Let E_m keep exactly the terms with $1 \leq |h| \leq m$. Since $s \geq 4m$, these h give distinct columns different from ℓ , so E_m is supported in the cyclic band; a band pair that wraps past s has its row in $\mathcal{B}$, and one that wraps past 0 has its column in $\mathcal{B}$, so the band lies in $\mathcal{P}$. By Lemma 5, $(E_m)_\Lambda$ has row sums at most ρ_2 , and as $\alpha^2 m^2 \geq w$ and $\pi \log_2 e > 1$, $\|(E - E_m)_\Lambda\| < 3e^{-\pi w} < 3 \cdot 2^{-w}$.

Entries. For $1 \leq |h| \leq m$ the exponent of a term of E_m is $\pi j/k$ with $k = s^2$ and the positive integer $j = \alpha^2((th + \sigma_\ell)^2 - \sigma_{\ell+h}^2)$, where $\sigma_\ell = s\beta_\ell = t\ell - sq_\ell$ is an integer of $O(p)$ bits, and $j > 0$ by part (c) of the proof of Lemma 5. [4, Lemma 2.13], applied with precision parameter w (its Section 2.2 fixes an arbitrary precision parameter larger than 100), returns an approximation of the real number x on the grid $2^{-w}\mathbb{Z}[i]$ with error less than $2 \cdot 2^{-w}$ in time $O(w^{1+\delta})$; its real part $\tilde{x} \in 2^{-w}\mathbb{Z}$ satisfies $|\tilde{x} - x| < 2 \cdot 2^{-w}$. Let $\tilde{A} = I + \tilde{E}_m$ be the matrix of these values. Then $\|(\tilde{E}_m - E_m)_\Lambda\| < 4m2^{L-w}$.

Elimination. Put $a^{(0)} = \tilde{A}$. For $k = 0, \dots, s-1$ let $\pi_k = a_{kk}^{(k)}$, $\tilde{u}_{kj} = a_{kj}^{(k)}$ for $j \in J_k$, $\psi_k = Q_w(1/\pi_k)$ and $\tilde{l}_{ik} = Q_w(a_{ik}^{(k)}/\pi_k)$ for $i \in J_k$, and for $i, j \in J_k$ set $a_{ij}^{(k+1)} = Q_w(a_{ij}^{(k)} - \tilde{l}_{ik}\tilde{u}_{kj})$, the product and difference being exact; leave all other entries unchanged. Put $\tilde{l}_{kk} = 1$, $\tilde{u}_{kk} = \pi_k$, and let every other factor entry be zero. If $(i, k), (k, j) \in \mathcal{P}$ and $i, j > k$, then $(i, j) \in \mathcal{P}$: if $k \in \mathcal{B}$ then $i, j \in \mathcal{B}$, and otherwise i or j lies in $\mathcal{B}$, or $i, j \in (k, k + m]$. Thus J_k lists exactly the possible nonzero positions, and nothing outside $\mathcal{P}$ is ever written.

Let $\varepsilon_{ij}^{(r)}$ be the truncation error of the update of (i, j) at step r , or zero, and for $i > j$ let $\varepsilon'_{ij} = \pi_j \tilde{l}_{ij} - a_{ij}^{(j)}$. Then $|\varepsilon_{ij}^{(r)}| < 2^{-w}$ and $|\varepsilon'_{ij}| < |\pi_j| 2^{-w}$, and telescoping the updates gives

$$\tilde{L}\tilde{U} = \tilde{A} + \Delta, \quad \Delta_{ij} = \sum_{r < \min(i, j)} \varepsilon_{ij}^{(r)} + [i > j] \varepsilon'_{ij},$$

with $\tilde{L} = (\tilde{l}_{ij})$ and $\tilde{U} = (\tilde{u}_{ij})$. For $k \leq s$ let $\Delta^{(k)}$ keep only the terms with $r < k$ and the ε'_{ij} with $j < k$, and put $M^{(k)} = \tilde{A} + \Delta^{(k)}$. The same telescoping shows that the computed $\tilde{l}_{ir}, \tilde{u}_{rj}$, $r < k$, satisfy the Crout equations of $M^{(k)}$ in all positions with $\min(i, j) < k$, and that $(a_{ij}^{(k)})_{i, j \geq k}$ is the exact k -th Schur complement of $M^{(k)}$.

Suppose $|\pi_r| \leq \frac{11}{10}$ for $r < k$. Then $|\Delta_{ij}^{(k)}| < (s+1)2^{-w}$, and since a row has at most s entries and $s < 2^p$, $\|\Delta_{\Lambda}^{(k)}\| < 2^L s(s+1)2^{-w} < 2^{2p+L-w} = 2^{-p-L-12}$. Hence $M_{\Lambda}^{(k)}$ satisfies Lemma 4 with $\rho \leq \rho_2 + 4m2^{L-w} + 2^{-p-L-12} < \frac{1}{10}$. Pivots are invariant under the diagonal similarity, so $\pi_k \in [\frac{9}{10}, \frac{11}{10}]$; the unscaled active entries satisfy $|a_{ij}^{(k)}| \leq \frac{11}{10}2^L$ and $|\tilde{l}_{ik}| \leq 2^L/9 + 2^{-w}$. This closes the induction. All stored values therefore fit in words of $L+3$ integer and sign bits and w fractional bits. At the end, $(\tilde{L}_{\Lambda}, \tilde{U}_{\Lambda})$ is the LU factorization of M_{Λ} , where $M = \tilde{A} + \Delta$, and it satisfies the bounds of Lemma 4. Also $|\psi_k - 1/\pi_k| < 2^{-w}$.

Factor stream and precomputation cost. The record F_k consists of ψ_k , the values $\tilde{u}_{kj}$ for $j \in J_k$, and the values $\tilde{l}_{ik}$ for $i \in J_k$, in increasing index order and followed by a separator: $O(m)$ words, hence $O(smw)$ bits in all. At step k only rows and columns in $W_k = \{k\} \cup J_k$ are read or written. An active entry outside $W_k \times W_k$ is an unchanged entry of $\tilde{A}$, because a row or column index larger than $k+m$ and outside $\mathcal{B}$ is in $\mathcal{P}$ -relation with no earlier pivot. Keep the at most $(2m+1)^2$ words of $W_k \times W_k$ row-major on a work tape. A step performs at most $(2m+1)^2$ products and divisions of $O(w)$ -bit numbers by schoolbook arithmetic, rewrites the window once, appends the new row and column with the $O(m)$ new entries of $\tilde{A}$, and writes F_k . The integers σ_{ℓ} come from the counters $tj = sf_j + e_j$ of the selected coordinates, since $\sigma_{\ell} = e_{\ell} - s \mathbf{1}_{2e_{\ell} \geq s}$. This proves the first claim.

Solve. Convert the input records to the fixed-point words and put $v = u/2$, which is exact. Forward: for $k = 0, \dots, s-1$ let $\tilde{y}_k = Q_w(\text{acc}_k)$, where $\text{acc}_k = v_k - \sum_{r < k} \tilde{l}_{kr} \tilde{y}_r$ is accumulated exactly. Keep the exact accumulators of the rows in $(k, k+m] \setminus \mathcal{B}$ in a queue on one tape and those of the rows in $\mathcal{B}$ on another, initialized with the corresponding v_i ; after $\tilde{y}_k$ is formed, one forward scan of F_k alongside both tapes subtracts $\tilde{l}_{ik} \tilde{y}_k$ for every $i \in J_k$, the head row leaves the queue, and the next row enters it. Backward: for $i = s-1, \dots, 0$ form exactly $g_i = \tilde{y}_i - \sum_{j \in J_i} \tilde{u}_{ij} \tilde{x}_j$ and $\tilde{x}_i = Q_w(\psi_i g_i)$, reading F_i backward record by record; the values $\tilde{x}_j$ with $j \in (i, i+m] \setminus \mathcal{B}$ are kept in a queue, and those with $j \in \mathcal{B}$, computed first, on a separate tape. Output $\tilde{J}u = Q_p(\tilde{x}/2^{L+1})$ in the scalar component format.

Errors. Forward, $\tilde{L}\tilde{y} = v + \xi$ with $|\xi_k| < \sqrt{2}2^{-w}$. Scaling gives $\tilde{L}_{\Lambda}\tilde{y}_{\Lambda} = v_{\Lambda} + \xi_{\Lambda}$ with $\|\xi_{\Lambda}\| < \sqrt{2}2^{-w}$ and $\|v_{\Lambda}\| \leq \frac{1}{2}$, hence $\|\tilde{y}_{\Lambda} - \tilde{L}_{\Lambda}^{-1}v_{\Lambda}\| \leq \frac{8}{7}\sqrt{2}2^{-w}$ and $\|\tilde{y}_{\Lambda}\| < 0.58$. Backward, downward induction gives $|\tilde{x}_j| \leq \lambda_j$: if this holds for $j > i$, then $|g_i| \leq (0.58 + \frac{1}{10})\lambda_i$ and $|\tilde{x}_i| \leq (\frac{10}{9} + 2^{-w})0.68\lambda_i < \lambda_i$. The residual $\eta_i = \tilde{x}_i - g_i/\pi_i$ satisfies $|\eta_i|/\lambda_i < (\sqrt{2} + 0.68)2^{-w} < 2.2 \cdot 2^{-w}$, and $\tilde{U}\tilde{x} = \tilde{y} + D_U\eta$. Lemma 4 for the scaled factors gives

$$\|\tilde{x}_{\Lambda} - M_{\Lambda}^{-1}v_{\Lambda}\| \leq \frac{5}{4} \cdot \frac{8}{7}\sqrt{2}2^{-w} + \frac{9}{8} \cdot 2.2 \cdot 2^{-w} < 5 \cdot 2^{-w},$$

hence $\|\tilde{x} - M^{-1}v\| < 5 \cdot 2^{L-w}$. Further, $\|N_{\Lambda} - M_{\Lambda}\| < 3 \cdot 2^{-w} + 4m2^{L-w} + 2^{-p-L-12} < 2^{-p-L-11}$, $\|N_{\Lambda}^{-1}\| \leq 1.004$ and $\|M_{\Lambda}^{-1}\| \leq \frac{10}{9}$, so

$$\|(N^{-1} - M^{-1})v\| \leq 2^L \cdot 1.004 \cdot \frac{10}{9} \cdot 2^{-p-L-11} \cdot \frac{1}{2} < 0.56 \cdot 2^{-p-11},$$

and $\|\tilde{x} - N^{-1}v\| < 2^{-p-10}$. Since $N^{-1}v/2^{L+1} = Ju$, the output error is less than $\sqrt{2} + 2^{-11} < 2$ units of 2^{-p} . Also $\|J\| \leq 1.004R/2^{L+2} \leq 0.251$, so the truncated output has modulus less than one and lies in $\mathbb{D}_p^s$.

Every exact accumulator is bounded by $\frac{1}{2} + 0.58 \cdot 2^L/8$ and every g_i by $0.68 \cdot 2^L$, so all words have $O(w)$ bits; as $L \leq p$, $w = O(p)$.

Solve cost. Each index uses $O(m)$ products of $O(w)$ -bit integers, $O(w^{1+\delta})$ each with the established multiplier, and $O(mw)$ head moves for one record of the factor stream and one pass over each queue or the $\mathcal{B}$ tape. The factor stream is traversed once forward and once backward. Initializing the $\mathcal{B}$ accumulators, reversing the output order and the record conversions take $O(sw)$. Hence the solve costs $O(sm w^{1+\delta})$ on a fixed number of tapes. $\square$

3.2 The revised one-dimensional interface

Proposition 2. *Fix $0 < \delta < 1/8$. Let $p > 100$, $2 \leq s < t < 2^p$ and $2 \leq \alpha < \sqrt{p}$ be integers with $\gcd(s, t) = 1$ and $t < 2s$, let L, w, m be as in Lemma 6, and assume $L \leq p$ and $s \geq 4m$. With $(P_s u)_j = u_{tj}$ and $(P_t v)_k = v_{-sk}$, the maps $A = S/2$ and $B_0 = D'JC$, where $D' = 2^{-(2\alpha^2-2)}D$, are contractions and*

$$P_s F_s = 2^{2\alpha^2+L+1} B_0 P_t F_t A.$$

There are fixed multitape algorithms for approximations $\tilde{A}$ and $\tilde{B}_0$ with errors less than p^2 and times $O(tp^{3/2+\delta}\alpha)$ and $O(tm w^{1+\delta} + tp^{1+\delta})$, the second given the factor stream of Lemma 6.

Proof. By Lemma 5 the matrix N is invertible, so the left-inverse identity $(DN^{-1}C)T = I$ of [4, (4.2)] holds, and the resampling identity [4, Theorem 4.2] gives $P_s F_s = DN^{-1}C P_t F_t S$. Inserting $D = 2^{2\alpha^2-2}D'$, $N^{-1} = 2^{L+2}J$ and $S = 2A$ proves the identity. The norms $\|A\| < \frac{3}{4}$ and $\|D'\| < 1$ hold as in the proof of [4, Proposition 4.7(i)], and $\|J\| \leq 0.251$ by Lemma 6. The approximations are $\tilde{A} = \tilde{S}'$ and $\tilde{B}_0 = \tilde{D}'JC$, with the cited algorithms for S' (error $< 16p$) and D' (error < 4), the banded solve (error < 2), and the exact ordered selection C . Errors add through the contractions, so $\varepsilon(\tilde{B}_0) < 4 + 2 < p^2$ and $\varepsilon(\tilde{A}) < 16p < p^2$.

The cited statements are used outside the hypothesis $\theta > p/\alpha^4$ of their Proposition 4.7. We record the sentences of their proofs that are used, by section and printed page of the cited manuscript; none of them involves θ .

- Section 4.1, p. 24: “let s and $t > s$ be positive integers such that $\gcd(s, t) = 1$, and let $\alpha \in (0, \infty)$ ”; Theorem 4.2, p. 24: “We have $TP_s F_s = P_t F_t S$ ”.
- Lemma 4.5, p. 26: “We have $\|S\| < 1 + \alpha^{-1}$.” The estimate (4.1), p. 27: “ $\|D\| \leq e^{\pi\alpha^2/4}$ ”. The left inverse (4.2), p. 28: “ $(DN^{-1}C)T = DN^{-1}T' = D(T'D)^{-1}T' = I$ ”, valid whenever $N = T'D$ is invertible.
- Proof of Proposition 4.7(i), p. 29: “Lemma 4.5 implies that $\|S\| < \frac{3}{2}$ ” and “By (4.1) we have $\|D\| \leq e^{\pi\alpha^2/4} < 2^{1.14\alpha^2} < 2^{2\alpha^2-2}$, as $1.14x < 2x - 2$ for all $x > 4$; hence also $\|D'\| < 1$ ”. The last inequality holds for every $x > 2.33$, in particular for $x = \alpha^2 = 4$.
- Lemma 4.8, pp. 29–30, computing D' : its proof uses that “ α, s, t and ℓ are all integers with $O(p)$ bits (here we have used the hypotheses $s, t < 2^p$ and $\alpha < p^{1/2}$)”, the estimate $\|D'\| < 1$, and Lemma 2.14 “with $\sigma := 2\alpha^2 - 2 < 2p$ ”.
- Lemma 4.9, pp. 30–31, computing S' with $m = \lceil p^{1/2} \rceil \alpha$ terms: its proof uses “Since $\|u\| \leq 1$ and $\alpha > 2$ we have”, only to bound the weight $\frac{1}{2}\alpha^{-1}$ by $\frac{1}{4}$, which also holds for $\alpha = 2$; then “Since $\alpha < p^{1/2}$ we have $w^m = e^{-\pi \lceil p^{1/2} \rceil / \alpha} \leq e^{-\pi p^{1/2} / \alpha} < e^{-\pi} < 0.05$ ”, “As $m < \lceil p^{1/2} \rceil p^{1/2} \leq p + p^{1/2}$

and $p > 100$ ”, and “Recalling from the proof of Proposition 4.7(i) that $\|S'\| < \frac{3}{4}$ ”. Its statement assumes the hypotheses of Proposition 4.7, which allow $2 \leq \alpha < p^{1/2}$.

- Section 2.2, p. 8: “Fix a precision parameter $p > 100$ ”; Lemma 2.13, p. 17, is stated for this arbitrary precision and is applied above with precision w .

The cited Lemmas 4.11 and 4.12 (the Neumann solve), and Lemma 4.6, are not used. $\square$

For the tensor interface, every axis i uses its own L_i , the scale becomes $\gamma = \sum_i (2\alpha^2 + L_i + 1)$, and the factor stream of axis i is written once on a dedicated tape before its lines are compressed. The error bound dp^2 and the contraction properties are unchanged. The one-dimensional calls cost $\sum_i (T/t_i) O(t_i p^{3/2+\delta} \alpha + t_i m_i w_i^{1+\delta})$, and the precomputations $O(\sum_i s_i m_i^2 w_i^2)$.

4 Parameters, costs and the witness

Keep $b = \lceil \log_2 n \rceil$, $p = 6b$, $d = \lfloor b^\epsilon \rfloor$, $K = d$, $T, r, \ell, t_i, \eta = 1/(4d)$ and the prime selection $(1 - 2\eta)t_i < s_i \leq (1 - \eta)t_i$ of the pinned source. Set

$$\alpha = 2, \quad L_i = \left\lceil \frac{4533 s_i}{4000 (t_i - s_i)} \right\rceil, \quad \gamma = \sum_{i=1}^d (2\alpha^2 + L_i + 1).$$

Then $\theta_i > \eta$ gives $s_i/(t_i - s_i) < 4d$, so $L_i \leq \lceil 4.533d \rceil$, and for $d \geq 22$

$$\gamma \leq d(10 + 4.533d) \leq 5d^2 \leq 5b^{2\epsilon}.$$

Also $s_i > (1 - 2\eta)t_i \geq t_i/2$, $L_i \leq p$, $w_i \leq 6p$, $m_i \leq \sqrt{6p}$ and $s_i \geq r/4 \geq 4m_i$ for large n , so Proposition 2 applies on every axis. Every later use of γ in the pinned precision analysis needs only $\gamma \leq b/4$, which holds once $b^{1-2\epsilon} \geq 20$. The Gaussian line maps cost $O(dTp^{3/2+\delta})$ in total, and the precomputations $O(drp^3) = o(Tp)$ because $T/r \geq (r/2)^{d-1}$ and r exceeds every fixed power of p . After division by Tp , the Gaussian row has power $\frac{1}{2} + \delta + \epsilon$. No other row of the cost table changes. The seven margins are

$$\begin{aligned} g_1 &= 1 - \epsilon(1 + c), & g_2 &= \epsilon c a_b, \\ g_3 &= \epsilon(1 - \lambda'), & g_4 &= a_b(1 - \epsilon), \\ g_5 &= \frac{1}{2} - \delta - \epsilon, & g_6 &= 1 - \delta - \epsilon, & g_7 &= \epsilon. \end{aligned}$$

Relative to the baseline constraint list, exactly five entries change: the guard $1 - \epsilon(5 - 4\beta + \zeta)$ becomes $1 - \epsilon$; the Gaussian cost $\frac{1}{4} - \delta - \frac{5}{4}\epsilon$ becomes $\frac{1}{2} - \delta - \epsilon$; the dimension and source-scale conditions $\frac{1}{3} - \epsilon$ and $\frac{1}{2} - \frac{3}{2}\epsilon$ become $\frac{1}{2} - \epsilon$ and $1 - 2\epsilon$; and $\alpha < \sqrt{p}$ becomes unconditional.

Put $t = 10^{-12}$ and choose

$$\begin{aligned} \tau &= 1 - a_b, \quad \sigma = 1 - a_c, \quad \beta = \frac{1}{10}, \quad c = 1, \quad \delta = \frac{1}{10^{12}}, \quad \epsilon = \frac{124999999}{250000000}, \\ \lambda &= 1 - a_b(1 - t/2), \quad \lambda' = 1 - a_b(1 - t), \quad \kappa = \frac{296461013}{2 \cdot 10^{17}}. \end{aligned}$$

Since $\sigma < \tau$, the compact-control recurrence has $\chi = \tau + (1 - \beta) \max\{\sigma - \tau, 0\} = \tau$, and exact comparisons give $\max\{\tau, \sigma, \chi\} < \lambda < \lambda' < 1$, $\sigma + \beta(1 - \sigma) < \lambda'$ and $1 - c = 0 < \lambda'$. All the other strict conditions hold, with $1 - 2\epsilon = 8/10^9$. The minimum margin is

$$\begin{aligned} G_* &= g_3 = \epsilon a_b(1 - t) = \frac{3705762684806317521666189976715649}{25 \cdot 10^{41}}, \\ G_* - \kappa &= \frac{22306317521666189976715649}{25 \cdot 10^{41}} > 0, \end{aligned}$$

while $g_5 = 3999/10^{12}$, $g_1 = 8/10^9$ and $g_2, g_4 > G_*$. The strict gap absorbs all fixed powers of $\log p$.

Theorem 1 (Conditional refinement). *Under the retained upstream, network and compact-control interfaces, and the written proofs of this note,*

$$T(n) = O\left(n(\log n)^{1-296461013/(2 \cdot 10^{17})}\right).$$

The saving lies strictly between 2^{-30} and 2^{-29} and exceeds the previous witness $5929220328/10^{19}$ by the factor $7411525325/2964610164 \approx 2.49999997$.

The choice of ϵ is within $4 \cdot 10^{-9}$ of $\frac{1}{2}$. The resulting eventual cutoffs, for example $b \geq 2^{541000000}$ for $\gamma \leq b/4$, are astronomically large; they are fixed constants, which the asymptotic statement allows. Any fixed $\epsilon < \frac{1}{2}$ gives a witness with smaller cutoffs; $\epsilon = \frac{2}{5}$ gives about twice the previous witness with $b^{1/5} \geq 20$.

5 The supremum and the ceiling of this assembly

Write $q = 1 - \lambda'$. For the current savings ($\sigma < \tau$) the layer conditions can be met exactly when $q < \min(a_b, (1 - \beta)a_c)$, and β now enters no other condition. With $c = 1$, the binding margins are $g_3 = \epsilon q$ and $g_5 = \frac{1}{2} - \delta - \epsilon$; their crossing $\epsilon \rightarrow 1/(2 + 2q)$ lies strictly below every other cap. Hence the parameter supremum is

$$\kappa_{\text{sup}} = \frac{Q}{2 + 2Q}, \quad Q = \min(a_b, a_c),$$

about $1.4823050814 \cdot 10^{-9}$ for the current savings. The previous assembly gives $Q/(5 + 4Q)$.

The value is within the factor $1 + Q$ of a ceiling that no Gaussian stage can remove. The retained conditions include $g_2 = \epsilon c a_b > \kappa$, $g_3 = \epsilon q > \kappa$, the transform layout requirement $K = \lfloor d^c \rfloor \leq \ell - 1$, which gives $\epsilon(1 + c) < 1$, and the prime-interval requirement $\log(r/2)/d \rightarrow \infty$, which gives $1 - 2\epsilon > 0$. Therefore

$$\kappa < \epsilon \min(q, c a_b) \leq \frac{\min(q, c a_b)}{\max(1 + c, 2)} \leq \frac{q}{2} \leq \frac{Q}{2}.$$

Exceeding $Q/2$ requires relaxing both $\epsilon < \frac{1}{2}$ caps: the chunk layout of the synthetic transform and the prime-interval argument. Further gains otherwise come only from the finite networks: after the linear guard, β may be arbitrarily small, so improving a_b alone raises κ proportionally until a_b reaches a_c .

6 Reproduction and verification boundary

The implementation uses only Python's standard library:

```
python3 scripts/assembly_lu.py
python3 scripts/make_assembly_lu_patch.py
python3 -m unittest discover -s tests -p 'test_assembly_lu*.py' -v
git apply --check --directory=upstream patches/assembly-lu-29.patch
```

The certificate re-derives a_b with the retained paired-bit certificate and a_c from the pair-star counts and a rational logarithm enclosure. It recomputes the node allowance D_{node} and C_0^* from the pair-star circuit, the entries of $C^{\otimes f}$ for $f \leq 12$, rational enclosures of π , $e^{2\pi}$, $e^{4\pi}$, $e^{14\pi}$ and $\log 2$, every constant of Lemmas 4–6, the identities for ψ on exact samples, and both scaled-exponent inequalities exactly on every row of six instances. It checks every strict constraint and margin of the witness, and that the retained constraint list agrees with the repository's code. The tests run the factorization

and solve of Lemma 6 in exact arithmetic at $p = 101$, including two $\alpha = 2$ instances whose unscaled off-diagonal row sums exceed 0.46, and compare the output with a reference solve of the full matrix at more than 200 digits; the worst observed error is below 1.4 units of 2^{-p} .

The new patch applies to the unmodified pinned manuscript. It incorporates the earlier compact-control and pair-star modifications, replaces the guard subsection, inserts the banded solve before the one-dimensional resampling lemma and revises that lemma, and updates the assembly parameters, cost table, margins and witness. The verifier does not implement the multiplication machine, and the finite tests do not prove uniform tape estimates or the cited analytic identities. Those remain stated proof dependencies. No unrestricted optimality, independent review, or practical performance claim is made.

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